Assignment - 1

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Answers

45/50

Task-1: Design of Learning systems.

System	Learning Type	Task
Loan default	supervised	classification
Customer segmentation	unsupervised	clustering
Credit-card spending	supervised	Regression
ATM cash refill	Reinforcement	decision-making
Fraud detection	semi-supervised	classification.

Conclusion:

The semi-supervised fraud detection system (e) can be highly affected by 2% of the records are labelled.

Task-2: Bayes Theorem - Disease Predi...

given;

Disease prevalence = 2% = 0.02

sensitivity = 95% = 0.95

specificity = 90% = 0.90

False positive rate = 0.10.

using Bayes' theorem:

$$P(D/+) = \frac{P(+/D)P(D)}{P(+)}$$

$$= \frac{0.95 \times 0.02}{(0.95 \times 0.02) + (0.10 \times 0.98)}$$

$$= \frac{0.019}{0.117}$$

$$\therefore \boxed{P(D/+) = 0.1624 = 16.24\%}$$

Task-3: Joint probability distribution:

Given;

X/Y	Y = 0	Y = 1
X = 0	0.10	0.20
X = 1	0.15	0.25
X = 2	0.20	0.10

a] marginal probabilities.

For X: $P(X=0) = 0.10 + 0.20 = 0.30$

$P(X=1) = 0.15 + 0.25 = 0.40$

$P(X=2) = 0.20 + 0.10 = 0.30$

For Y:

$$P(Y=0) = 0.10 + 0.15 + 0.20 = 0.45$$

$$P(Y=1) = 0.20 + 0.25 + 0.10 = 0.55$$

b] Expected values & variance:

$$E[X] = \sum x\,p(x)$$

$$= 0(0.3) + 1(0.4) + 2(0.3)$$

$$= 1,$$

$$E[X^2] = 0 + 0.4 + 4(0.3) = 1.6$$

$$\text{var}(X) = E[X^2] - (E[X])^2$$

$$= 1.6 - 1^2 = 0.6.$$

$$\boxed{E[Y] = 0(0.45) + 1(0.55) = 0.55}$$

c] Conditional probability:

$$P(X=2/Y=1) = \frac{P(X=2, Y=1)}{P(Y=1)}$$

$$= \frac{0.10}{0.55} = 0.1818.$$

d] Covariance & correlation:

$$E[XY] = 0.25 + 0.20 = 0.45$$

$$\text{cov}(X,Y) = E[XY] - E[X]E[Y]$$

$$= 0.45 - (1 \times 0.55)$$

$$\boxed{\text{cov}(X,Y) = -0.10.}$$

$$\text{var}(Y) = 0.55(0.45) = 0.2475$$

$$r = \frac{\text{cov}(X,Y)}{\sqrt{\text{var}(X)\,\text{var}(Y)}}$$

$$= \frac{-0.10}{\sqrt{0.6 \times 0.2475}} \Rightarrow \boxed{r = -0.2595}$$

c) Independence:

X & Y are not Independent because:

$$P(X=0, Y=0) = 0.10$$

but,

$$P(X=0)P(Y=0) = 0.30 \times 0.45 = 0.135$$

∴ since both values are not equal, X and Y are not independent.

Task-4: Normal Distribution and maximum Likelihood Estimation:

Given;

$$X \sim N(50, 16)$$

$$\mu = 50, \quad \sigma = 4$$

a] Calculate probabilities:

For $P(X > 58)$

$$Z = \frac{58 - 50}{4} = 2$$

$$P(X > 58) = 1 - \phi(2)$$

$$= 1 - 0.9772$$

$$\boxed{P(X > 58) = 0.0228}$$

For $P(46 < X < 54)$

$$Z_1 = \frac{46 - 50}{4} = -1$$

$$Z_2 = \frac{54 - 50}{4} = 1$$

$$P(-1 < Z < 1) = 0.8413 - 0.1587$$

$$\boxed{P = 0.6826}$$

b) Maximum Likelihood Estimation:

given readings:

12, 15, 11, 14, 13

mean: $\mu = \frac{12+15+11+14+13}{5}$

$$\boxed{\mu = 13}$$

ML variance:

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{n}$$

$$= \frac{1+4+4+1+0}{5}$$

$$\boxed{\sigma^2_{ML} = 2}$$

∴ unbiased variance:

$$s^2 = 10/5-1$$

$$\boxed{s^2 = 2.5}$$

Task-5:- Information Theory – Entropy:

given: $P(1/2, 1/4, 1/8, 1/8)$

$$\therefore H(x) = -\sum p(x) \log_2 p(x)$$

$$= -\left[\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{4}\log_2\frac{1}{4} + \frac{1}{8}\log_2\frac{1}{8} + \frac{1}{8}\log_2\frac{1}{8}\right]$$

$$= 0.5 + 0.5 + 0.375 + 0.375$$

$$\boxed{H(x) = 1.75 \text{ bits}}$$

uniform source Entropy:

$$H(x) = \log_2 4 = 2 \text{ bits.}$$

e) Independence:

X & Y are

Average code length:

given, 0, 10, 110, 111

$L = \sum P(x) L(x)$

$= (0.5 \times 1) + (0.25 \times 2) + (0.125 \times 3) + (0.125 \times 3)$

$L = 1.75$ bits

∴ Conclusion:- The code is optimal because its average code length is equal to entropy.

Task-6: Kullback-Leibeler Divergence

given: $P(0.5, 0.25, 0.25)$

$Q = (1/3, 1/3, 1/3)$

a] $KL(P||Q)$

$D_{KL}(P||Q) = \sum P(x) \log_2 \frac{P(x)}{Q(x)}$

$= 0.085$ bits.

b] $KL(Q||P)$

$= 1/3 [\log_2(2/3) + \log_2(4/3) + \log_2(4/3$

$D_{KL}(Q||P) = 0.087$ bits.

∴ $D_{KL}(P||Q) \neq D_{KL}(Q||P)$.

Task-1: Entropy and Mutual Information.

given:

$P(0,0) = 0.4$, $P(0,1) = 0.1$

$P(1,0) = 0.1$, $P(1,1) = 0.4$

M.P:

$P(A=0) = 0.5$, $P(A=1) = 0.5$

$P(B=0) = 0.5$, $P(B=1) = 0.5$

a) Entropy of A and B:

$$H(A) = -(0.5\log_2 0.5 + 0.5\log_2 0.5)$$

$$= 1 \text{ bit} //$$

similarly, $H(B) = 1 \text{ bit} //$

b) Joint Entropy:

$$H(A,B) = -\sum P(A,B)\log_2 P(A,B)$$

$$= -[2(0.4\log_2 0.4) + 2(0.1\log_2 0.1)]$$

$$\boxed{H(A,B) = 1.7219 \text{ bits}}$$

c] Conditional Entropy:

$$H(A/B) = H(A,B) - H(B)$$

$$= 1.7219 - 1 = 0.7219 \text{ bits} //$$

d) mutual Information:

$$I(A;B) = H(A) - H(A/B)$$

$$= 1 - 0.7219$$

$$= 0.2781 \text{ bits.} //$$

Task-8: Bayes' Decision Theory:

given:

$$P(x/\omega_1) = N(2,4)$$

$$P(x/\omega_2) = N(6,4)$$

$$P(\omega_1) = 0.6, \quad P(\omega_2) = 0.4$$

a) Bayes Decision Threshold:

$$x^* = \frac{\mu_1 + \mu_2}{2} + \frac{\sigma^2}{\mu_2 - \mu_1} \ln \frac{P(\omega_1)}{P(\omega_2)}$$

Substituting:

$$x^* = \frac{2+6}{2} + \frac{4}{6-2} \ln \frac{0.6}{0.4}$$

$$= 4 + \ln(1.5) \Rightarrow \boxed{x^* = 4.4055}$$

b] Classification of $x = 4.5$

Since:

$$4.5 > 4.4055$$

∴ the observation belongs to: $\boxed{\omega_2}$

c) overall prob. of Error:

using normal distribution:

$$= 0.6(0.11454) + 0.4(0.21265)$$

$$\boxed{P(\text{error}) \approx 0.1538 = 15.38\%}$$

d] Effect of prior prob.

If the prior prob. of ω_1 increases, the decision threshold moves towards the mean of ω_2.

This makes classification into ω_1 more likely.

Task-9: minimum Risk decision:

given: $P(\omega_1/x) = 0.3$, $P(\omega_2/x) = 0.7$

Loss matrix:

Action	Disease	Healthy
Declare disease	0	8
Declare healthy	50	0

a) Conditional Risk of α_1

$$R(\alpha_1) = 0(0.3) + 8(0.7)$$

$$= 5.6.$$

b] α_2)

$$R(\alpha_2) = 50(0.3) + 0(0.7)$$

$$= 15$$

c] mini. Risk Action:

Since:

$$5.6 < 15$$

The mini-Risk action is

α_1 = Declare disease.

d] Comparision with MAP Decision:

MAP selects healthy because 0.7 is greater than 0.3.

Posterior Threshold:

2

$$8(1-P) < 50P$$

$$8 < 58P$$

$$P > 0.13793?$$

Correction:

$$P > 8/58 = 0.13793. \Rightarrow 13.79\%.$$

Task-10: Polynomial Regression -

Bias & variance Trade-off.

~~given~~:

a] Generalisation Gap.

Gap = validation MSE - Training MSE

Degree	Calc.	Gap
1	9.5 - 9.0	0.5
2	4.6 - 4.1	0.5
3	2.9 - 2.2	0.7
5	3.6 - 0.9	2.7
9	11.8 - 0.1	11.7

b] Degree-1: underfitting & errors are high

Degree-2: Better than degree 1

Degree-3: Good balance b/w train & validation errors.

Degree-5: shows sign of overfitting.

Degree-9: severe overfitting & train error is very Low & validation error is high.

c] Degree 3 is the best model because it has the lowest validation MSE of 2.9.

d]. 1. Regularization: Use Ridge or Lasso regression to reduce model complexity.

2. more Training Data: Increase the training dataset to improve generalisation.

Task-11: ...gesian

given: n=10, h=-

prior: Beta(2,2)

a] MLE: $\hat{\theta}_{ML}$

b] MAP:

Beta(2+7

MAP = 9-1 / 9+

c] Posteri... mea...

d] If T...

TASK-1...

a]. De...

1. Ta...

2. Pe...

3. E...

Task-11: Bayesian Estimation:

given. $n = 10, h = 7, t = 3$

prior:

Beta(2,2)

a) MLE:

$$\hat{\theta}_{ML} = 7/10 = 0.7$$

b) MAP:

$$Beta(2+7, 2+3) = Beta(9,5)$$

$$MAP = \frac{9-1}{9+5-2} = 0.6667.$$

c) Posterior mean:

$$mean = 9/9+5$$

$$= 0.6429.$$ //

d) If 70 heads occurs in 100 tosses:

$$ML = 70/100 = 0.7$$

$$MAP = 71/102 = 0.6961$$

$$p.m = 72/104 = 0.6923.$$ //

Task-12:

a]. Design of Learning System:

1. Task (T):

2. Performance (P):

3. Experience (E):

4. Target function:

5. Input Representation.

6. Learning Algorithm.

7. validation strategy.

b] Calc. TP, FN, FP & TN.

given, Total = 1000

sepsis prevalence = 5%.

sensitivity = 90%.

specificity = 80%.

TP: $TP = 50 \times 0.90 = 45$ %

FN: $FN = 50 - 45 = 5$ %

TN: $TN = 950 \times 0.80 = 760$ %

FP: $FP = 950 - 760 = 190$ %

c] $PPV = \frac{TP}{TP + FP} = 19.15\%$.

$NPV = 760/760 + 5$

$= 99.35\%$.

d] given: cost of missed sepsis = ₹100

cost of false alarm = ₹5

$= (5 \times 100) + (190 \times 5)$

$= 500 + 950$

∴ Total = ₹1,450

∴ Threshold = 4.76%.